

A Kronecker preconditioner for the EFGP normal equations

1 The system being preconditioned

In the EFGP gradient path (see `efgpnd.py::efgpnd_gradient_batched`), each hyper step requires solving

$$A\beta = b, \quad A = DTD + \sigma^2 I \in \mathbb{C}^{M \times M}, \quad (1)$$

where $M = m^d$ is the number of Fourier nodes on a d -dimensional tensor grid, $D = \text{diag}(w)$ with $w_j = \sqrt{\hat{k}(\xi_j)} h^d$ encoding the spectral density of the (stationary, product) kernel, and $T \in \mathbb{C}^{M \times M}$ is the block-Toeplitz “data Gram” whose entries are $T_{ij} = \sum_{n=1}^N e^{2\pi i(\xi_i - \xi_j) \cdot x_n}$. The solver is PCG with tolerance 10^{-4} .

Exact matrix assembly costs $\mathcal{O}(M^2)$; neither that nor $\mathcal{O}(M^3)$ factorisation is an option when M reaches $10^3 10^5$. An application $v \mapsto Av$ runs in $\mathcal{O}(d M \log m)$ via an FFT-based Toeplitz product. PCG’s runtime is therefore $\mathcal{O}(k_{\text{iter}} \cdot d M \log m)$, so the practical question is *how fast does k_{iter} grow*, and what can the preconditioner do about it.

2 Two easy baselines and why they can fail

Jacobi ($\text{diag}(A)^{-1}$). The diagonal of A is exactly $T_0 |w_j|^2 + \sigma^2$, where $T_0 = v_{\text{kernel}}[0] \approx N$. Jacobi is essentially free and is adequate when the cross-terms of DTD are not dominant — i.e. a high-noise, moderate- ℓ , well-spread-sample regime. It degrades quickly as ℓ shrinks or $\sigma^2 \rightarrow 0$ in $d \geq 2$: the near-null space of DTD becomes large, the tail frequencies have tiny $|w_j|^2$, and the scaled operator still has condition number $\mathcal{O}(\kappa(T))$.

Randomised Nyström. Form $P = U\Lambda U^\top + \mu(I - UU^\top)$ with $\mu = \sigma^2$ and (U, Λ) a rank- k Nyström approximation of $A - \sigma^2 I$ (`create_nystrom_precond`). Excellent when A has a clear top- k eigengap (clustered samples, anisotropic geometry); essentially useless otherwise, because for uniform spreads the spectrum of DTD decays smoothly and no small rank is enough. Setup also costs k full A -applies, so it is only worth it when the resulting iteration count is small.

3 The Kronecker preconditioner

3.1 Structure being exploited

For a product-separable spectral density $\hat{k}(\xi) = \prod_{j=1}^d \hat{k}_j(\xi_j)$ (e.g. the squared exponential, product Matérn), the weight vector factorises on the tensor grid:

$$w[i_1, \dots, i_d] = \prod_{k=1}^d D_k[i_k], \quad D_k[i_k] = \sqrt{\hat{k}_k(\xi_{i_k})} h \quad \text{up to a normalising constant.} \quad (2)$$

The data Toeplitz T is characterised by its symbol $v(\delta) = \sum_{n=1}^N e^{2\pi i \delta \cdot x_n}$, or equivalently by the empirical characteristic function $\hat{\varphi}_N(\delta) := v(\delta)/N$. If the samples are drawn iid from a *product measure* $p(x) = \prod_{k=1}^d p_k(x_k)$, the true CF factorises, $\varphi_X(\delta) = \prod_k \varphi_{X_k}(\delta_k)$, and by the law of large numbers so does the empirical one:

$$\hat{\varphi}_N(\delta) \longrightarrow \prod_{k=1}^d \hat{\varphi}_{N,k}(\delta_k) \quad \text{as } N \rightarrow \infty, \quad \hat{\varphi}_{N,k}(\delta_k) := \frac{1}{N} v(0, \dots, \delta_k, \dots, 0), \quad (3)$$

with finite-sample error $\mathcal{O}(1/\sqrt{N})$. Pulling out the N 's, this is $T \approx \bigotimes_{k=1}^d T_k$ in expectation, where the T_k are 1-D Toeplitz blocks built from the axis- k slices of v . Writing $H_k := D_k T_k D_k$ and $H := \bigotimes_k H_k$,

$$A \approx H + \sigma^2 I, \quad H = \bigotimes_{k=1}^d H_k. \quad (4)$$

A deterministic grid is the zero-error limit; any iid product measure (uniform, independent Gaussian, iid mixtures, etc.) works too, with error shrinking as N grows. Since each $H_k = V_k \Lambda_k V_k^H$ is Hermitian positive semidefinite (size $m \times m$), the full eigendecomposition of H is

$$H = \left(\bigotimes_k V_k \right) \Lambda \left(\bigotimes_k V_k \right)^H, \quad \Lambda = \bigotimes_k \Lambda_k, \quad (5)$$

and A^{-1} applies in $\mathcal{O}(d M m)$ flops.

3.2 Why $T \approx \bigotimes_k T_k$: a finite-sample statement

Empirical characteristic functions. Let $X = (X_1, \dots, X_d) \sim p$ with $p(x) = \prod_{k=1}^d p_k(x_k)$, and let $\{x_n\}_{n=1}^N \stackrel{\text{iid}}{\sim} p$. Write

$$\hat{\varphi}_N(\delta) := \frac{1}{N} \sum_{n=1}^N e^{2\pi i \delta \cdot x_n}, \quad \hat{\varphi}_{N,k}(\delta_k) := \hat{\varphi}_N(\delta_k e_k) = \frac{1}{N} \sum_{n=1}^N e^{2\pi i \delta_k x_{n,k}}, \quad (6)$$

for the joint and axis- k marginal empirical CFs. Let $\varphi(\delta) := \mathbb{E}[e^{2\pi i \delta \cdot X}]$ and $\varphi_k(\delta_k) := \mathbb{E}[e^{2\pi i \delta_k X_k}]$; independence gives $\varphi(\delta) = \prod_k \varphi_k(\delta_k)$, with $|\varphi_k| \leq 1$.

For any compact $K \subset \mathbb{R}^d$,

$$\sup_{\delta \in K} \left| \hat{\varphi}_N(\delta) - \prod_{k=1}^d \hat{\varphi}_{N,k}(\delta_k) \right| = O_p(N^{-1/2}).$$

Proof. Define the (complex) empirical processes

$$W_N(\delta) := \sqrt{N}(\hat{\varphi}_N(\delta) - \varphi(\delta)), \quad W_{N,k}(\delta_k) := \sqrt{N}(\hat{\varphi}_{N,k}(\delta_k) - \varphi_k(\delta_k)).$$

Each is a normalised average of iid bounded complex random variables (modulus ≤ 1), so by the multivariate CLT plus standard equicontinuity arguments for empirical CFs (e.g. Csörgő 1981; the class $\{e^{2\pi i \delta \cdot x} : \delta \in K\}$ is Donsker), each converges in distribution in $\ell^\infty(K)$ to a mean-zero Gaussian process and is therefore $O_p(1)$ uniformly on K .

Substitute $\hat{\varphi}_{N,k}(\delta_k) = \varphi_k(\delta_k) + N^{-1/2}W_{N,k}(\delta_k)$ and expand:

$$\prod_{k=1}^d \hat{\varphi}_{N,k}(\delta_k) = \prod_k \varphi_k(\delta_k) + N^{-1/2} \sum_k \varphi_{\setminus k}(\delta) W_{N,k}(\delta_k) + N^{-1} \mathcal{R}_N(\delta), \quad (7)$$

where $\varphi_{\setminus k}(\delta) := \prod_{j \neq k} \varphi_j(\delta_j)$ (hence $|\varphi_{\setminus k}| \leq 1$), and $\mathcal{R}_N$ collects all cross-products of two or more $W_{N,k}$'s. Since each $W_{N,k}$ is $O_p(1)$ uniformly on K , so is $\mathcal{R}_N$.

Subtracting (7) from $\hat{\varphi}_N(\delta) = \varphi(\delta) + N^{-1/2}W_N(\delta)$ and using $\varphi(\delta) = \prod_k \varphi_k(\delta_k)$:

$$\hat{\varphi}_N(\delta) - \prod_k \hat{\varphi}_{N,k}(\delta_k) = N^{-1/2} \left[W_N(\delta) - \sum_k \varphi_{\setminus k}(\delta) W_{N,k}(\delta_k) \right] + O_p(N^{-1}),$$

and the bracket is $O_p(1)$ uniformly on K . $\square$

Identification of T and $\bigotimes_k T_k$. The Gram is exactly N times the joint empirical CF evaluated on frequency differences:

$$T_{ij} = \sum_{n=1}^N e^{2\pi i(\xi_i - \xi_j) \cdot x_n} = N \hat{\varphi}_N(\xi_i - \xi_j). \quad (8)$$

For the axis- k Toeplitz, the code assembles a 1-D kernel vector from the axis- k slice $v(\delta_k e_k)$ of v_{kernel} and applies the normalisation $T_k \leftarrow \tilde{T}_k / v_{\text{kernel}}[0]^{(d-1)/d}$ with $v_{\text{kernel}}[0] = N$. Unpacking:

$$\tilde{T}_k[i_k, j_k] = v((\xi_{i_k, k} - \xi_{j_k, k}) e_k) = N \hat{\varphi}_{N,k}(\xi_{i_k, k} - \xi_{j_k, k}),$$

so after normalisation

$$(T_k)_{i_k, j_k} = N^{1/d} \hat{\varphi}_{N,k}(\xi_{i_k, k} - \xi_{j_k, k}). \quad (9)$$

Taking the Kronecker product and collapsing $N^{1/d}$ factors:

$$\left(\bigotimes_{k=1}^d T_k \right)_{i,j} = \prod_k (T_k)_{i_k, j_k} = N \prod_{k=1}^d \hat{\varphi}_{N,k}(\xi_{i_k, k} - \xi_{j_k, k}). \quad (10)$$

The $(d-1)/d$ exponent is *exactly* the normalisation that makes the N 's collapse correctly, so both T and $\bigotimes_k T_k$ are N times a characteristic function — the joint CF in (8), and its coordinate-wise factorisation in (10).

Entrywise rate. Subtracting and applying Lemma 3.2 with $K = \{\xi_i - \xi_j : i, j \in [m]^d\}$ (finite, hence compact):

$$R_{ij} := T_{ij} - \left(\bigotimes_k T_k \right)_{ij} = N [\hat{\varphi}_N(\delta_{ij}) - \prod_k \hat{\varphi}_{N,k}(\delta_{ij, k})] = O_p(\sqrt{N}), \quad (11)$$

uniformly in (i, j) , where $\delta_{ij} := \xi_i - \xi_j$. Contrast with $T_{ii} = N$: relative entrywise error is $O_p(N^{-1/2})$.

Operator norm. A crude but sufficient bound uses Frobenius: $\|R\|_{\text{op}} \leq \|R\|_F = (\sum_{ij} |R_{ij}|^2)^{1/2} = O_p(\sqrt{N}M)$. Meanwhile $\|T\|_{\text{op}} \geq \max_j T_{jj} = N$, so for fixed M (i.e. fixed quadrature accuracy ε and fixed ℓ), the relative operator error $\|R\|_{\text{op}}/\|T\|_{\text{op}} = O_p(M/\sqrt{N}) \rightarrow 0$ as $N \rightarrow \infty$. Tighter bounds via non-commutative concentration (Tropp matrix Bernstein, applied to $R = \sum_n R^{(n)}$ with each $R^{(n)}$ the contribution of sample x_n) give $\|R\|_{\text{op}} = O_p(\sqrt{N \log M})$, shaving the M to a $\sqrt{\log M}$, but (11) is the content of the approximation.

Sandwiching by D and comparing to $A = DTD + \sigma^2 I$, the preconditioned operator $M^{-1/2}AM^{-1/2} = I + M^{-1/2}(DRD)M^{-1/2}$ has

$$\|M^{-1/2}(DRD)M^{-1/2}\|_{\text{op}} = O_p(N^{-1/2}) \quad (\text{at fixed } M, \ell, \sigma^2 > 0),$$

whence $\kappa(M^{-1/2}AM^{-1/2}) = 1 + O_p(N^{-1/2})$ and CG attains tolerance ε in $O(\log \varepsilon^{-1})$ steps independent of N . The tensor-grid case is the degenerate $N^{-1/2} \rightarrow 0$ limit where $R \equiv 0$ deterministically.

3.3 What we actually build

For general (non-gridded) samples, T is *not* exactly separable, but its Kronecker restriction remains a useful approximation: the operator $\tilde{A} := (\bigotimes_k H_k) + \sigma^2 I$ uses, as T_k , the 1-D Toeplitz whose symbol is the axis- k marginal of v_{kernel} (i.e. the slice with all other indices pinned at 0) and the Jacobi-style normalisation $D_k \leftarrow D_k/w_{\text{ctr}}^{(d-1)/d}$, $T_k \leftarrow T_k/v_{\text{kernel}}[0]^{(d-1)/d}$, so that $\bigotimes_k H_k$ reproduces $A - \sigma^2 I$ exactly on tensor-grid data (and to leading order on well-spread data).

The preconditioner is then

$$M^{-1} = \left(\bigotimes_k V_k \right) (\Lambda + \sigma^2 I)^{-1} \left(\bigotimes_k V_k \right)^H. \quad (12)$$

The apply is three passes over the d -dimensional tensor: (i) mode- k multiplication by V_k^H for $k = 1, \dots, d$; (ii) pointwise division by the Kronecker-diagonal $(\prod_k \lambda_k^{(i_k)}) + \sigma^2$; (iii) mode- k multiplication by V_k . No $M \times M$ matrix is ever materialised.

4 Complexity

	Setup (one-time)	Apply (per PCG iter)
Jacobi	$\mathcal{O}(M)$	$\mathcal{O}(M)$
Nyström (rank r)	$\mathcal{O}(r \cdot dM \log m)$ (r A-applies) + $\mathcal{O}(Mr^2)$	$\mathcal{O}(Mr)$
Kronecker	$\mathcal{O}(dm^3)$	$\mathcal{O}(dMm)$
A-apply (reference)	—	$\mathcal{O}(dM \log m)$

The per-iteration cost of applying the Kronecker preconditioner is *larger* than one A -apply by a factor of $m/\log m$. For $m = 39$ this is $\sim 11\times$. Kronecker is therefore only a net win if it reduces the PCG iteration count by at least that factor.

Setup is a one-time $\mathcal{O}(dm^3)$. In $d \geq 3$ this is *cheaper* than a single A -apply. In $d = 2$ it equals roughly ten A -applies, which is negligible once it is amortised over a handful of CG iterations or Adam steps.

5 When it helps, theoretically

Consider the spectrum of A . Write the eigenvalues of each H_k as $\lambda_k^{(i_k)}$. The Kronecker-preconditioned operator is

$$M^{-1/2}AM^{-1/2} = I + M^{-1/2}\left(T - \bigotimes_k T_k\right)D \cdot D M^{-1/2}.$$

The condition number is governed by the *non-separable remainder* $R := T - \bigotimes_k T_k$. Three regimes:

(a) Exactly separable data. If samples lie on a tensor grid, $R = 0$ and $M^{-1/2}AM^{-1/2} = I$. PCG converges in exactly one step. Our `grid` benchmark confirms this (1 iter vs. 2000+ unpreconditioned).

(b) Well-spread, non-gridded data. For generic $\{x_n\}$ in $[-L, L]^d$ with bounded second moment, R is a *low-order* perturbation of the separable tensor — roughly, it captures only the cross-axis fluctuations that 1-D marginals miss. In this regime $\|M^{-1/2}RM^{-1/2}\| \ll 1$ and PCG converges in $\mathcal{O}(\log \epsilon^{-1})$ iterations, independent of N , ℓ , and σ^2 to leading order. Empirically we see 3–40 iterations on uniform SE in $d = 2$ across the entire range of (ℓ, σ^2) we tested, including the Jacobi-killing short- ℓ /low- σ corner.

(c) Strongly clustered data. When samples concentrate near p points (“ p -cluster” data), T acquires a rank- p structure that is *not* captured by any product of 1-D marginals, because the cross-axis correlation is exactly what the clustering induces. Here R is large and Kronecker degrades to roughly the same iteration count as Jacobi; Nyström is the appropriate tool because a rank- $\mathcal{O}(p)$ correction resolves the structure.

6 Summary

Data geometry	Spectrum of R	Best preconditioner
Tensor grid	$R = 0$	Kronecker (1 iter)
Jittered grid / Poisson / uniform	low rank + fast decay	Kronecker
Mildly anisotropic	similar	Kronecker
p -cluster, Gaussian blob	rank- p , not separable	Nyström (rank $\gtrsim p$)
Extreme high noise ($\sigma^2 \gg \ w\ _\infty^2 T_0$)	R dominated by $\sigma^2 I$	Jacobi suffices

Algorithmic conditions under which Kronecker is the right default. All three of:

1. *Separable kernel:* $\hat{k}(\xi) = \prod_k \hat{k}_k(\xi_k)$. Holds for SE, product-Matérn, product-rational-quadratic.
2. *Non-clustered samples:* empirical density of x_n is bounded above and below on its support. Uniform, jittered-grid, and low-variance log-Gaussian covariates all qualify. Point clusters do not.
3. *Moderate d :* the per-iter overhead $m/\log m$ is small. For $d \leq 3$ and quadrature $\epsilon = 10^{-4}$ giving $m \leq 40$, this is always $\leq 11\times$, and the iteration savings are much larger than that.

When these hold simultaneously, Kronecker is roughly 1.5–2 $\times$ faster wall-clock than Jacobi on warm Adam trajectories, and 10–100 $\times$ faster at the stiff corners that Jacobi fails to traverse ($\ell \ll 1$, $\sigma^2 \ll 1$). Cost of being wrong (“always default to Kronecker, including on clustered data”) is a

small overhead ($\sim 2\times$ slower than Jacobi in the worst non-ideal case), which is why it is a safe default in the EFGP pipeline for any product kernel.